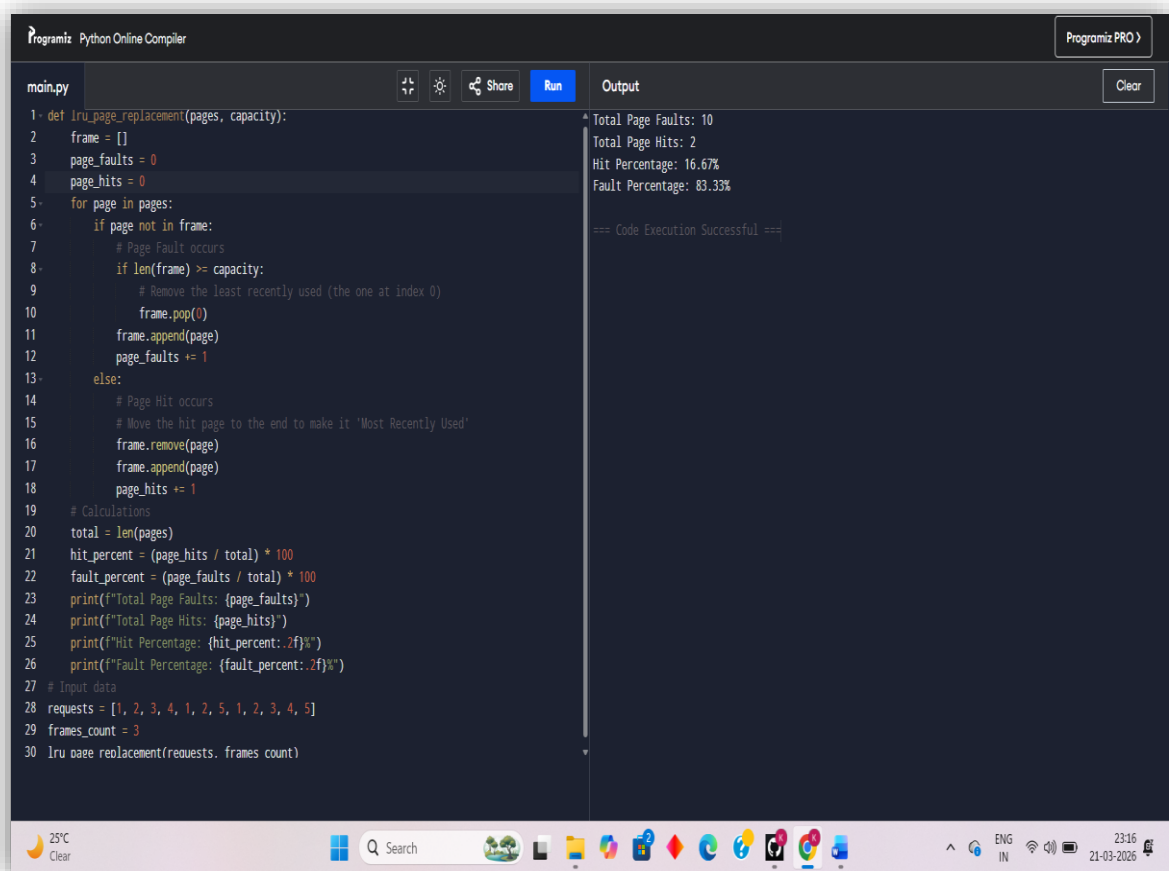


A System has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).



```
main.py
1- def lru_page_replacement(pages, capacity):
2     frame = []
3     page_faults = 0
4     page_hits = 0
5     for page in pages:
6         if page not in frame:
7             # Page Fault occurs
8             if len(frame) >= capacity:
9                 # Remove the least recently used (the one at index 0)
10                frame.pop(0)
11            frame.append(page)
12            page_faults += 1
13        else:
14            # Page Hit occurs
15            # Move the hit page to the end to make it 'Most Recently Used'
16            frame.remove(page)
17            frame.append(page)
18            page_hits += 1
19    # Calculations
20    total = len(pages)
21    hit_percent = (page_hits / total) * 100
22    fault_percent = (page_faults / total) * 100
23    print(f"Total Page Faults: {page_faults}")
24    print(f"Total Page Hits: {page_hits}")
25    print(f"Hit Percentage: {hit_percent:.2f}%")
26    print(f"Fault Percentage: {fault_percent:.2f}%")
27    # Input data
28    requests = [1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5]
29    frames_count = 3
30    lru_page_replacement(requests, frames_count)
```

Output

```
Total Page Faults: 10
Total Page Hits: 2
Hit Percentage: 16.67%
Fault Percentage: 83.33%

=== Code Execution Successful ===
```

25°C Clear

Q Search

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